Increasing PV Hosting Capacity by using Voltage Support of Smart Inverters in MEA's LV Distribution Systems

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***Abstract*—High penetration of grid-connected photovoltaic (PV) systems in low voltage (LV) distribution system can pose challenges such as voltage rise issue. To prevent such problems, Metropolitan Electricity Authority (MEA) limits the amount of installed capacity of PV systems, PV hosting capacity, connected to the LV grid. This may not support the growth of renewable energy. Nowadays, smart inverters are designed with voltage support functions which can help stabilize voltage levels on the grid. This paper presents an increase in PV hosting capacity by using volt-var mode of smart inverters, which is one of voltage support functions, in MEA's LV distribution system. Optimal volt–var curve setting of smart inverters for using in MEA's LV distribution system will be investigated in this paper and DIgSILENT simulation will be also conducted by using MEA's typical LV system model. The results show that volt-var mode of smart inverters help mitigate voltage rise issues caused by high PV penetration and allows for more PV systems to be integrated into MEA's LV distribution systems without voltage rise issue.**

Keywords—smart inverter, volt-var mode, PV hosting capacity

I. Introduction

Over the last few years, the installation of PV systems has experienced rapid growth, attributed to the decrease in the price of PV systems, and the development of novel PV technologies. Currently, Thailand is undergoing a transition to clean energy era to achieve the target of carbon neutrality by 2050. Solar PV is expected to be one of pivotal players in response to such target. Additionally, Thailand's Alternative Development Energy Plan (AEDP) for 2018 to 2037 has established a target of 12,139 MW for PV power generation by 2037 [1].

One of the main limitations for further PV integration into existing LV networks is the voltage rise caused by the reverse power flow along the distribution feeders, especially during periods of high PV generation and low load consumption. High R/X ratio, which characterizes of distribution networks, causes overvoltage and reverse power flow due to significant active power injections associated with the incorporation of PV systems on the distribution networks. To prevent voltage rise issue, MEA limits the amount of installed capacity of PV systems connected to the LV grid, which should not exceed 15% of the rated power of the distribution transformer, light load during daylight hours. This limitation may not support Thailand's AEDP.

Several techniques to alleviate the voltage rise issue have been proposed. Many utilities may choose to curtail real power feed-in from PV units at times of low demand. Some utilities may choose to reinforce the distribution grid, by increasing conductor sizes to reduce the resistance of LV lines. Among many advanced techniques, the use of voltage support function of smart inverters is highly effective and economical to mitigate voltage rise and improve high PV penetration in LV networks. This paper presents an increase in PV hosting capacity by using the volt-var mode of smart inverter in MEA's LV distribution system. This feature allows smart inverter to manage reactive power output based on the local voltage, stabilizing the grid and mitigating issues caused by high PV penetration. Optimal volt–var curve setting of smart inverters for using in MEA's LV distribution systems will be investigated in this paper. DIgSILENT simulation conducted by using MEA's typical LV system model shows that smart inverters help mitigate voltage rise issues caused by high PV penetration and allows for more PV systems to be integrated into MEA's LV distributions system.

II. Preliminaries

A. Basic Theory

To understand how PV system can create the voltage rise over the power line, a simple circuit of a power system shown in Figure 1 will be used to explain here. This simplified power system consists of a generator on the left hand side, a power line, a load and a PV system on the right hand side [2].

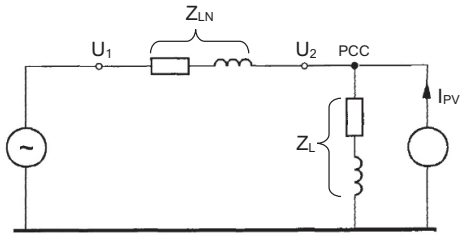


Fig. 1. Equivalent diagram of a simple radial power system

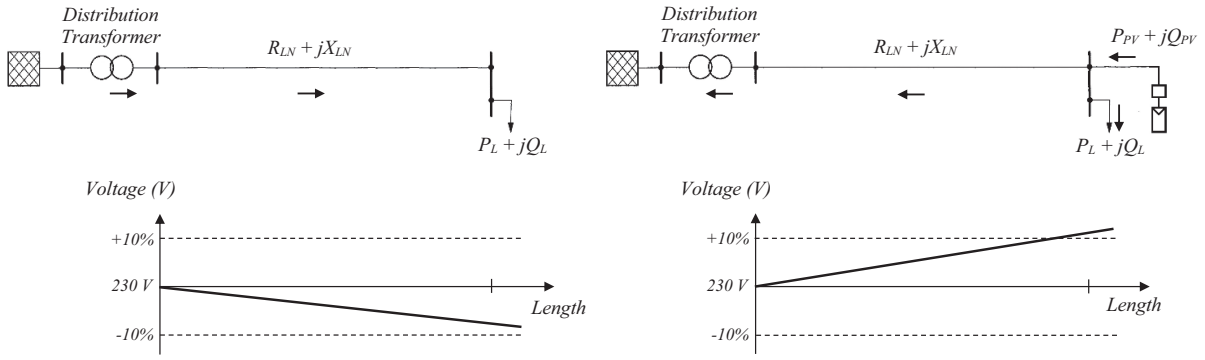
The grid voltage at point of common coupling (PCC) can be expressed approximately as (1).

$$U_2 = \frac{Z_L}{Z_{LN} + Z_L} \times (U_1 + Z_{LN} I_{PV}) \quad (1)$$

Where, U_1 is the source voltage, U_2 is the voltage at PCC, I_{PV} is feed-in current of grid-connected PV system, Z_{LN} and Z_L are line and load impedance, respectively.

It's clear that voltage without PV system ($I_{PV} = 0$) is lower than the source voltage. On the other hand, when PV system connected at the end of the circuit, voltage is increased proportional to I_{PV}. Therefore, the reverse power flow due to the PV system changing the normal voltage profile over the network length.

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(a) without PV system

(b) with PV system

Fig. 2. Voltage profile over the lines with and without grid connected PV system

Figure 2 (a) and (b) show voltage profile over the lines with and without grid connected PV system, respectively. Conventional design of power grid considers a load flow directed from transformer to the load. A qualitative voltage profile without PV system decreases over the network length. The voltage drop level depends on line impedance and the load condition. Normally, Utilities must keep voltage in the certain range as their standards as shown in Figure 2 (a) for propose of appliances and machinery operating properly.

When a PV system is connected at the end of the line as shown in Figure 2 (b), the load flow may change its direction. Especially, when the power generated by PV is higher than the energy consumed by load, the surplus electricity will flow back to the grid. A voltage profile with a PV system increases over the network length. Other devices and equipment might be damage, if the voltage rise at connection point of PV system exceeds the upper limit. Many electricity utilities limit the capacity (active power) of PV system that can be connected to a distribution system in order to prevent overvoltage. This may lead to the biggest barriers to mass distribution of PV system.

In [3], inverters with active power curtailment schemes are proposed to overcome the problem. These allow the inverters to inject maximum available power to the grids, as long as the ac bus voltage is below a certain value. Above this value, the injected power is reduced linearly with the ac bus voltage increase to keep voltage in normal level. However, a disadvantage is that the PV power output is dumped to control the voltage, leading to lower efficiency of the PV system. This can also lead to unfairness among users since the PV output at the end of the line tends to be restricted with higher priority.

Furthermore, there are many studies in [4] that can increase the capacity of PV system connected to a distribution system without overvoltage problem, such as network reinforcement, energy storage, active/reactive power control by PV inverter and using on load tap changer for MV/LV transformer etc. Each technique has its own advantages and disadvantages. In this paper, reactive power absorption by PV inverter is chosen and applied in LV distribution system to overcome the problem

B. Voltage control by Reactive Power

All Solar inverter can operate in all four quadrants [5]. It is able to inject and absorb reactive power while active power is fed into the grid. Reactive power absorption by PV inverters can decrease voltage rise over the line and increase the installed PV capacity. This can be explained below;

The voltage drop without PV system on the feeder is given by

$$\left|U_1 - U_2\right| = \left|I(R_{LN} + jX_{LN})\right| = \left|\frac{S_L{}^*}{U_2{}^*}(R_{LN} + jX_{LN})\right| \quad (2a)$$

$$= \left|\frac{(P_L - jQ_L)}{U_2}(R_{LN} + jX_{LN})\right| \quad (2b)$$

$$= \left|\frac{(R_{LN}P_L + X_{LN}Q_L) + j(X_{LN}P_L - R_{LN}Q_L)}{U_2}\right| \quad (2c)$$

For small power flow, the voltage angle between U_1 and U_2 in (2a) is small, and the voltage drop can be approximated by

$$\left|U_1 - U_2\right| \approx \left|\frac{R_{LN}P_L + X_{LN}Q_L}{U_2}\right| \quad (3)$$

For a system with load and PV system, the voltage drop on the feeder can be approximated by

$$\left|U_1 - U_2\right| \approx \left|\frac{R_{LN}(P_L - P_{PV}) + X_{LN}(Q_L - (\pm Q_{PV}))}{U_2}\right| \quad (4)$$

Where R_{LN} is line resistance, X_{LN} is line reactance, S_L P_L and Q_L are apparent, active and reactive power consumed by load, respectively. While P_{PV} is active power injected by PV system and $\pm Q_{PV}$ is reactive power injected or absorbed by PV system.

Equation (4) indicates that voltage drop depends on PV active and reactive power relative to the load active and reactive power and the equivalent *X/R* ratio of the line. If voltage at PCC exceeds the upper limit due to active power injected into the grid, reactive power absorption by PV inverter can help to decrease the voltage rise and surplus of solar generated active power can also be fed into the grid.

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The reactive power absorption methods can be divided mainly into four categories, namely, fixed power factors (fixed cos φ), fixed reactive power (fixed Q), reactive power as a function of voltage in the connection point ($Q(U)$) and power factor as a function of injected active power (cos $\varphi(P)$) [6]. In general, these methods also have the drawback that they can increase power loss. The reactive power absorption by the PV inverter causes a higher current of inverter. Due to the higher current, additional power losses are generated in the inverter and the grid lines, But the benefit is higher active power can be transmitted, and a surplus of solar generated electrical power can be fed into the grid.

C. Q(U) mode of Smart inverter

In many international standards, the smart inverter can provide voltage regulation capability by changes of reactive power [7]. The *Q(U)* mode is the most commonly used smart inverter function due to its effectiveness for voltage control by bidirectional reactive power injection and absorption. This feature allows smart inverters to manage reactive power output based on the local voltage, stabilizing the grid and mitigating issues caused by high PV penetration [8].

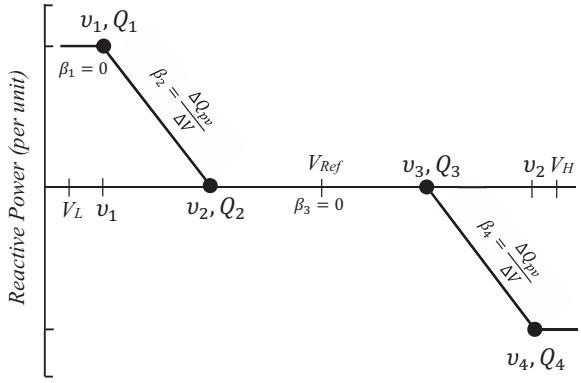


Fig. 3. Generic Q(U) curve and defining parameters.

A typical *Q(U)* curve is shown in Figure 3. The smart inverter injects its rated var (Q_1) when the voltage at PCC is between υ_1 and the low voltage V_L. The var injection is curtailed along the slope β_2 when the PCC voltage is between υ_1 and υ_2. Between υ_2 and υ_3, the dead-band is introduced where no vars are injected. With PCC voltage between υ_3 and υ_4, the var adsorption is curtailed along the slope β_4. At voltages beyond υ_4 to the high voltage V_H, the rated smart inverter's var is absorbed. The reactive power injection and absorption at the PCC as a function of the voltage can be expressed as (5)

$$Q(\upsilon_{pv}) = \begin{cases} Q_1 & \upsilon_{pv} \le \upsilon_1 \\ \beta_2 \times (\upsilon_{pv} - \upsilon_2) & \upsilon_1 < \upsilon_{pv} \le \upsilon_2 \\ 0 & \upsilon_2 < \upsilon_{pv} \le \upsilon_3 \\ \beta_4 \times (\upsilon_{pv} - \upsilon_3) & \upsilon_3 < \upsilon_{pv} \le \upsilon_4 \\ Q_4 & \upsilon_{pv} > \upsilon_4 \end{cases} \quad (5)$$

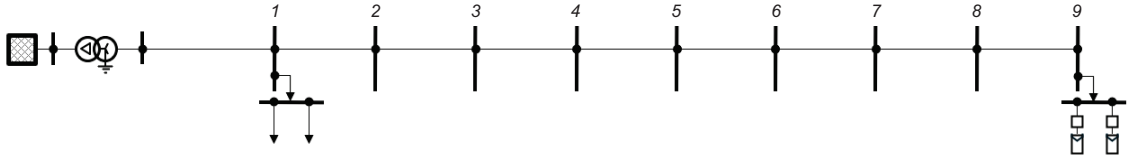
Where β_2 and β_4 is slope of *Q(U)* curve at reactive injection and absorption, respectively.

The *Q(U)* curve of smart inverters for use in MEA's LV distribution systems shows in Table I. Voltage set points refer to MEA's normal and emergency operating voltage range. Smart inverters shall be designed to inject and absorb reactive power before grid voltage is out of normal operating voltage range, which is between 0.93 and 1.03 per unit. At voltages beyond 1.03 per unit, the voltage-active power function is used to address the voltage rise caused by PV system, instead.

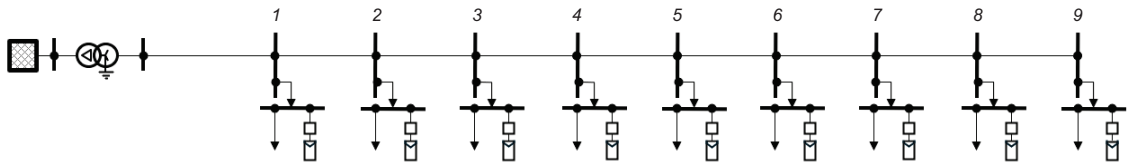
TABLE I. MEA'S VOLTAGE-REACTIVE POWER SETTINGS

Voltage-reactive power parameters	Setting
V_{Ref}	V_N
V_1	0.93 V_N
Q_1	30% of nameplate apparent power rating, injection
V_2	0.94 V_N
Q_2	0
V_3	1.01 V_N
Q_3	0
V_4	1.03 V_N
Q_4	30% of nameplate apparent power rating, absorption

Noted that the extent to which these functions can mitigate the voltage issues depend on several factors such as number of PV systems, their reactive and real power capabilities and their relative locations, and feeder characteristics, e.g., length, impedances, topology, load distribution.



(a) Scenario 1



(b) Scenario 2

Fig. 4. Single line diagram of MEA's LV feeder typical model

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III. Simulation and Results

A. Typical LV network model for simulation

The study case is implemented on MEA's generic LV network as shown in Figure 4. The grid contains 9 households with one LV feeder supplied by one 150 kVA transformer, the most used transformers in MEA's network. The feeder has a nominal voltage of 230 Volt and the farthest node from the distribution transformer is 500 meters. LV Feeder Conductor size is 70 mm^2. All consumer load in this study is 22.5 kVA with 0.85 power factor, which is 15% of the rated power of the distribution transformer. PV inverters has been modelled as controlled current sources.

The simulated study case is divided into two scenarios. Scenario 1 involves all loads placed near the distribution transformer and all PV systems placed at the end of the feeder as shown in Figure 4 (a). Scenario 2 involves all loads and PV systems distributed along the line as shown in Figure 4 (b).

Each study scenario includes two cases, case 1 represents the situation where the PV inverters have no *Q*(*U*) control algorithm implemented, and Case 2 represents the situation where the PV inverters have *Q*(*U*) control algorithm.

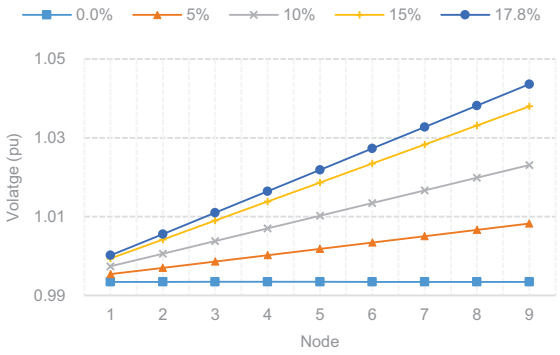
The upper and lower limit of MEA's service voltage is 1.043 per unit and 0.90 per unit respectively. All the scenarios are studied for variable PV penetration levels from zero to the highest possible PV penetration level, which the voltage level is not higher than the upper level. This provides numerous results to be analyzed, and a comparison between the study scenarios.

B. Simulation Results

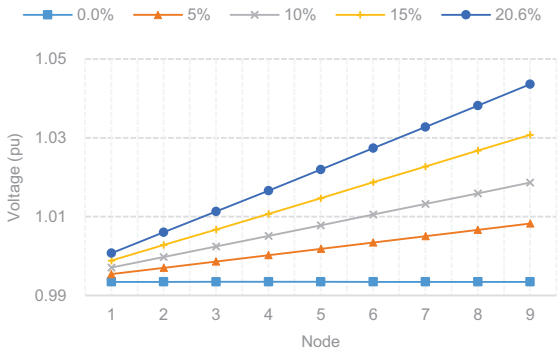
The voltage of LV side of the distribution transformer is defined as 1.0 per unit in no load condition. The distribution transformer can change tap position resulting in a 2.5 percent voltage increase and decrease at the secondary side, but for this study a constant tap position for transformer is used. Figure 5(a) and 6(a) display voltage levels at each node of feeder for variable PV penetration levels without *Q(U)*. Figure 5(a) and 6(a) display voltage levels at each node of feeder for variable PV penetration levels with *Q(U)*.

In scenario 1, all loads are placed near the transformer and all PV systems are placed at the end of the feeder. It is obvious that for the farthest distances to the transformer the highest voltage variations will be observed. Increasing PV penetration to 17% of the rated power of the transformer will result in a maximum voltage increase of approximately 1.043 per unit in node 9 as shown in Figure 5(a). When *Q*(*U*) control is adopted, it can be observed from Figure 5(b) that the voltage variations of modes have the same trends with no *Q(U)*control. The PV penetration can be increased up to 20%.

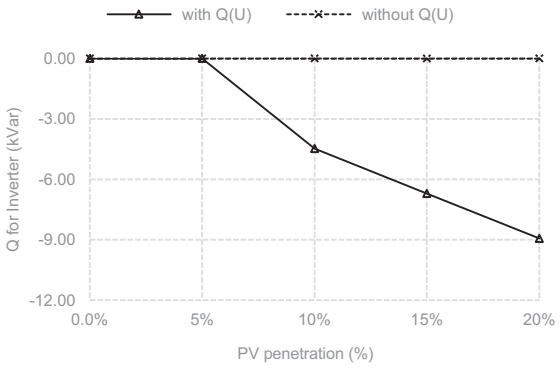
In scenario 2, all loads are distributed along the line and PV systems are placed near each load. In this scenario, hosting capacity before *Q(U)* adoption is 44% of the rated power of the transformer. When *Q*(*U*) is adopted, PV penetration can be increased up to 52% as shown in Figure 6(a) and 6(b), respectively.



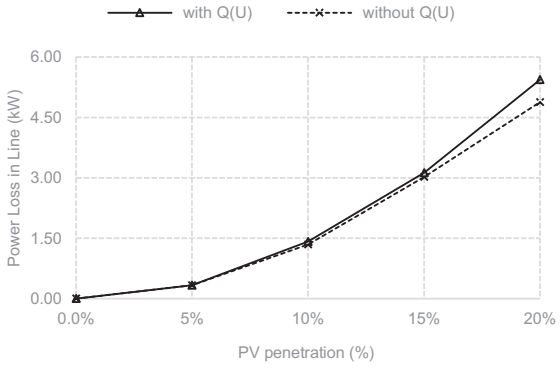
(a) Voltage at each mode for vaiable PV penetration without *Q(U)* mode



(b) Voltage at each mode for vaiable PV penetration with *Q(U)* mode



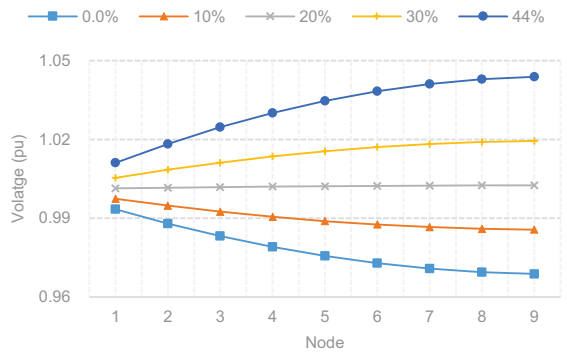
(c) Total reactive power from inverters for vaiable PV penetration with *Q(U)* mode and without *Q(U)* mode



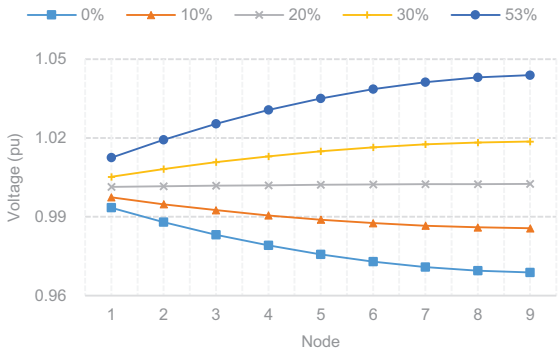
(d) Total line loss for vaiable PV penetration with *Q(U)* mode and without *Q(U)* mode

Fig. 5. Scenario 1 - loads placed near the transformer and PV systems placed at the end of the feeder

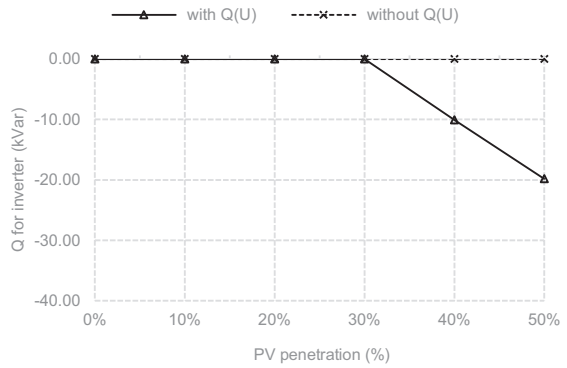
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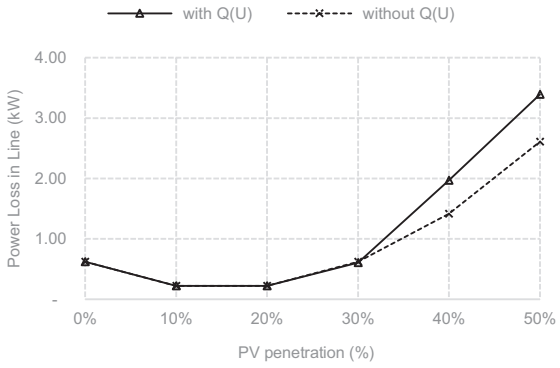
(a) Voltage at each mode for vaiable PV penetration without *Q(U)* mode



(b) Voltage at each mode for vaiable PV penetration with *Q(U)* mode



(c) Total reactive power from inverter for vaiable PV penetration with *Q(U)* mode and without *Q(U)* mode



(d) Total line loss for vaiable PV penetration with *Q(U)* mode and without *Q(U)* mode

Fig. 6. Scenario 2 - all loads and PV systems distributed along the line

TABLE II. COMPARISONS OF HOSTING CAPACITY FOR THE 2 SCENARIOS.

Scenario	PV Hosting Capacity (% of the rated power of the transformer)		Line loss increase
	Case 1 : without *Q(U)*	Case 2 : with *Q(U)*	
1	17%	20%	11.4%
2	44%	52%	30.0%

Table II shows the comparison of PV hosting capacity for 2 scenarios. The results obtained from simulation show that reactive power absorption by the smart inverter can increase PV hosting capacity in MEA's LV distribution system up to 20% and 52% of the rated power of the transformer in scenario 1 and 2, respectively. This depends on X/R ratio of the feeder. In case of 500 kVA Transformer and feeder size 185 mm^2 with a high X/R ratio, PV hosting capacity can be increased up to 31% of the rated power of the transformer in scenario 1.

Figure 5(c) and 6(c) displays total reactive power injected into the grid by inverters in different PV penetration and scenarios under *Q(U)* control. Smart inverters near the transformer are less effective in providing reactive power support to lower the maximum voltage due to the small voltage rise near the distribution transformer.

Figure 5(d) and 6(d) plots the power losses in lines for scenario 1 and 2, respectively. The situation with no voltage control at all in the network would generate a total power line loss of 0.62 kW for scenario 2. Power loss in line for scenario 1 is zero because of all loads connected near the transformer. There is no power flow through LV lines. For smaller PV penetration levels, there are no significant differences among the simulated cases, since the reactive power flow is also at minimum. With the PV penetration level increasing, different power losses can be observed due to the different reactive power absorption. The power losses in lines from *Q(U)* mode can increase by 11-30% depending on network topology and inverter settings.

IV. CONCLUSION

The volt-var mode in smart inverters can indeed increase the PV hosting capacity. By managing reactive power to maintain voltage levels, smart inverters help mitigate overvoltage issues caused by high PV penetration. This allows for more PV systems to be integrated into Utility's grid without compromising its stability.

This paper applied *Q(U)* mode of smart inverters in MEA's LV distribution network to increase PV hosting capacity. The results obtained from simulation show that reactive power absorption by the smart inverter can increase PV hosting capacity in MEA's LV distribution systems at least up to 20 percent of the rated power of the transformer. This depends on X/R ratio of the feeder. However, reactive power absorption can also lead to increased line losses up to

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11-30% due to the additional reactive power flow from smart inverters.

Accordingly, to mitigate overvoltage issues resulting from high levels of PV penetration, the volt-var or *Q(U)* mode along with additional grid support functionalities of advanced inverters, such as the volt-watt mode, shall be integrated into MEA's grid code and implemented as part of standard operational requirements.

REFERENCES

[1] Department of Alternative Energy Development and Efficiency, Ministry of Energy, "Thailand PV Status Report 2022," Bangkok, Thailand, 2022.

[2] A. Canova, L. Giaccone, F. Spertino and M. Tartaglia, "Electrical Impact Of Photovoltaic Plant In Distributed Network", in Industry Applications Conference, 2007. 42nd IAS Annual Meeting. Conference Record of the 2007 IEEE, 2007, pp. 1450-1455.

[3] IEA PVPS Task 10, report IEA-PVPS T10-06 (2009), "Overcoming PV grid issues in the urban areas".

[4] M. Vandenbergh, "D3.1 Prioritization of Technical Solutions Available for the Integration of PV into the Distribution Grid," Version 12, May 22, 2013.

[5] P. Esslinger and R. Witzmann, "Improving grid transmission capacity and voltage quality in low-voltage grids with a high proportion of distributed power plants," in Proc. Int. Conf. Smart Grid and Clean Energy (ICSGCE), Chengdu, China, Sep. 27–30, 2011.

[6] S. Hashemi, J. Østergaard, and G. Yang, "Effect of reactive power management of PV inverters on need for energy storage," in Proc. IEEE 39th Photovoltaic Specialists Conf. (PVSC), Tampa, FL, USA, Jun. 16–21, 2013, pp. 002153–002158.

[7] "IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces," IEEE Std 1547-2018, pp. 1–138, 2018.

[8] H. Lee, J.-C. Kim, and S.-M. Cho, "Optimal volt–var curve setting of a smart inverter for improving its performance in a distribution system," IEEE Access, vol. 8, pp. 157931–157945, 2020.

[9] T. O. Olowu, A. Inaolaji, A. Sarwat, and S. Paudyal, "Optimal volt-var and volt-watt droop settings of smart inverters," in Proc. IEEE Green Technologies Conf. (GREENTECH), 2021, pp. 89–96.

[10] K. Luo and W. Shi, "Comparison of voltage control by inverters for improving the PV penetration in low voltage networks," IEEE Access, vol. 8, pp. 161488–161497, 2020.

[11] "IEEE Application Guide for IEEE Std 1547™‐2018, IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces," IEEE Std 1547-2023.

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